ENS3553/ENS6160 Signals and Systems

Laboratory Experiment 2

Frequency Response of Systems

Prerequisites:

- A good understanding of the lab manual available on Canvas.
- A thorough review of the frequency response and filtering action of LTI systems, as presented in Modules 7 and 8.
- A review of the *RC* circuit used as an example in Module 7.
- A review of the techniques used in the first laboratory experiment.

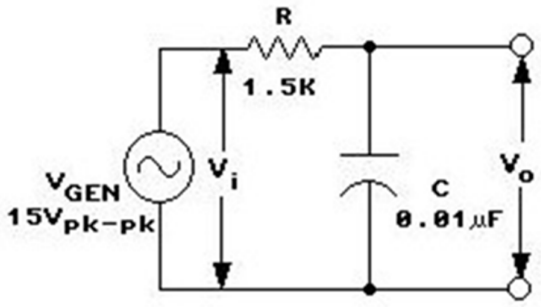
Objectives:

- To derive the frequency response of a filter and to use it to predict frequency response measurements.
- To measure the frequency response of CT LTI systems and to investigate the relationship between time response and frequency response characteristics.

Instructions/post lab

- Read the Lab Primer again carefully, review the first lab and make sure you are familiar with the techniques used in the first lab session. Refer to the oscilloscope guide documents provided to ensure you know how to operate it for this lab.
- Read carefully the lecture slides for Modules 6 and 7, along with the relevant sections of the textbook.
- Show your complete working for the following exercises in the lab results sheet, include all steps and be prepared to explain your answers to the laboratory supervisor:

A. <u>Determine and plot the frequency response of the RC circuit shown below:</u>



The circuit can be described by the linear constant-coefficient differential equation (LCCDE) $RC\frac{dV_c(t)}{dt} + V_c(t) = V_i(t)$.

a. Using the capacitor voltage V_o as the output and the source voltage V_i as the input, and using the resistor and capacitor values specified in the figure, determine the frequency response of this RC circuit.

   Note: Question 6 of tutorial 9 shows how to write the frequency response, $H(j\omega)$, of an LTI system using its LCCDE by setting the input to $e^{j\omega t}$. Use that approach and write an expression for the frequency response.

b. Work out $|H(j\omega)|$ – as an amplitude ratio ***and*** in dB - at frequencies 1 kHz, 10 kHz and 40kHz

c. Use a computer-based tool of your choice (e.g. Matlab) to plot the amplitude and phase response of the circuit as a function of frequency (from 0 to 60 kHz). Paste these plots in your lab result sheet. *Hint:* The sample Matlab code given in Module 6 can be modified to achieve this.

d. Based on your plot, answer the following question: What kind of filtering is performed by this RC circuit?

e. Identify the cut-off frequency.

   The "dividing line" between the passband and the stopband is the cut-off frequency, or ***-3dB point***. What does -3db equate to in terms of voltage amplitude ratio?

   Locate the -3dB point on your plot and note down the frequency.

   How can we determine this value from the circuit parameters?

f. Describe the behaviour of this system in the frequency domain and in the time domain.

B. <u>Determine the response of the RC circuit to an AC voltage by using the frequency response</u>

Module 5 covers how a sinusoidal input is modified by the frequency response of an LTI system.

a. Determine the output voltage of the RC circuit under investigation when the input is an AC voltage (i.e., sinusoid) of 15V peak-to-peak and phase shift 0°, for each of the following frequencies, *f*: 1 kHz, 10 kHz and 40 kHz.

b. Plot the magnitude and phase spectra of the resulting output voltages from the previous step.

C. Determine the response of the RC circuit to a square wave by using the frequency response

Module 5 covers how the frequency response of an LTI system relates the Fourier series coefficients of a periodic input to the Fourier series coefficients of the resulting output.

a. Write first 3 non-zero Fourier coefficients of a 15 V_{pk-pk} 8kHz square wave with 50% duty cycle. (Hint: Use the formula in post-lab for Lab 1).
b. Evaluate the magnitude of frequency response of the RC circuit (from Part A) at the frequencies of the non-zero harmonics evaluated in the previous step.
c. Hence work out the expected magnitude of the 3 harmonics at the output.
d. From the above results, describe what you think the output waveform will look like.

To save time in the lab, read the through the laboratory procedure, and use the results of the post-lab work to fill in the theoretical expected value sections of tables in the result sheet.

Laboratory Experiment

Equipment:

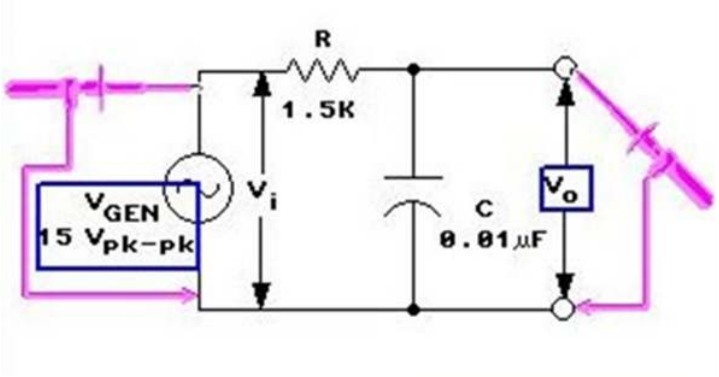
- Function generator (33220A)
- Digital Storage Oscilloscope (DSO-X 3024A)
- Bread board, 1 x 1.5kΩ resistor, and 1 x 0.01uF capacitor.

Procedure:

1. Verify the frequency response in the time domain by using the oscilloscope.

a. Use the bread board and supplied components to create the RC circuit illustrated in the post-lab. Connect the function generator across the input on your circuit. Adjust V_{GEN} for a 15 V_{pk-pk}, 1 kHz sine wave.

b. Turn on the oscilloscope, then connect the channel 1 (X) probe of the oscilloscope across the generator source and the channel 2 (Y) probe across the capacitor, as shown below. Display the two voltage waveforms on the oscilloscope screen. Use appropriate time and voltage scales.



c. Measure the output voltage magnitude and phase for each of the following input signal frequencies: 1 kHz, 10 kHz and 40 kHz. Complete Table A in the result sheet, comparing the theoretical frequency response to measured frequency response values. The theoretical values are provided by your results from the post-lab. The measured values can be obtained from the oscilloscope.

d. For each of the specified frequencies, sketch the input-output voltage pair you see on the oscilloscope screen in your result sheet.

e. Based on the measurements, does the output voltage increase or decrease with frequency? Does this confirm the type of filtering obtained in the post-lab?

f. Using measurements (by varying the input frequency), specify the approximate value of the cut-off frequency, or -3dB down point. How does it compare to the theoretical value obtained in the post-lab?

2. Verify the frequency response in the frequency domain by using the oscilloscope.

a. For each particular frequency, display first the frequency spectrum of the output on the oscilloscope and record its dBV magnitude, then do the same for the spectrum of the input. Complete Table B in the result sheet, highlighting the effect of the RC filter on the spectrum of the input sinusoid. Comment on the filtering effect.

b. Compare the magnitude response results with the theoretical values determined in the post-lab.

c. For each frequency specified, sketch the output spectra you got on the oscilloscope screen in your results sheet and compare it to the corresponding spectrum you got in the post-lab. Are they in good agreement?

3. Set the function generator for a 15 $V_{pk\text{-}pk}$, 8 kHz square wave output.

a. Following a similar approach as for the sine wave in 2, complete Table C in the result sheet. Use your post-lab results to calculate the theoretical value of the magnitude response at the kth harmonic $\omega = k\omega_0$ in dB.

b. Record in your result sheet the input and output spectra (on the same plot, to scale). Explain how different harmonics are affected by the frequency response.

c. Sketch the input and output time domain voltage waveforms on the same plot, to scale. Explain the shape of the output voltage waveform as a result of the filtering operation performed by the RC circuit.